Activity 4: Lithography-Etch Interaction Case

Course: Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Duration: 45 minutes

TSC alignment: A3

PPT alignment: Trainer Slides v7.1, concept slides 138-209 and Activity 4 overview slide 426.

Workplace scenario

Contact resistance rises after a lithography focus excursion and a longer-than-normal plasma over-etch.

Goal

Determine how variation propagates between lithography, etch, clean and contact modules.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Read the lot history and separate observations from assumptions.
2. Build an interaction matrix for resist thickness, focus, exposure, CD, etch bias, selectivity and residue.
3. Trace the most likely propagation path from lithography variation to contact resistance.
4. Identify the measurement that would discriminate between under-patterning, over-etch damage and post-etch residue.
5. Recommend an immediate containment action and one experiment that tests the suspected interaction.

Required evidence

Interaction matrix, causal chain and discriminating verification plan.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] At least three module interactions are identified

[ ] Containment protects downstream lots

[ ] The proposed test distinguishes competing causes

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.